



directing these to hit the target.

The second major concern is to produce sufficient damage to the intended target. This is where interaction of high energy with matter comes into play. This implies that, having a high-power laser or a HPM emitter alone does not make a DEW.

Three important constituents of DEW therefore are the high-energy sources influencing operational range, target-tracking and beam-pointing technology determining probability of target hit and interaction of high-energy beams with matter that determines lethality.

Types of DEWs

Four major categories of DEWs are:

1. PBWs
2. Microwave based DEWs
3. Laser based DEWs
4. LIPC weapons

A PBW uses a high-energy beam of atomic or subatomic particles to inflict intended damage to the target by disrupting its atomic and/or molecular structure. It is the least mature of the four DEW technologies and receives by far the least amount of research effort. It is not a true DEW. Unlike high-energy laser weapons and high-power microwaves that direct electromagnetic energy towards the target, it delivers kinetic energy into the target's atomic structure and is only a hard-kill weapon.

A microwave based DEW system is designed to produce the equivalent of electromagnetic interference to damage enemy's electronics systems. Due to concerns regarding unintended side effects on the host platform, it is usually preferred to put such weapons only on unmanned combat air vehicles. Also under consideration is use of high-power microwaves

as a weapon to attack underground and deeply-buried targets that are resistant to high explosives.

At the core of the laser based DEW is a high-power laser that has enough power in the case of continuous wave laser, or sufficient pulse energy in the case of pulsed laser, to inflict physical damage to the target. Though the lasers intended for already-established applications such as range finding,

airborne laser systems for destroying ballistic missiles and space based laser systems for neutralising theatre and inter-continental ballistic missiles.

A large number of experiments with laser based DEWs to demonstrate these or similar capabilities have been carried out in different parts of the world. Realisability of these weapons has been established beyond doubt, and these weapons have been projected by strategists as the weapons of the 21st century.

LIPC weapons are hybrid weapons that use a laser to ionise a path of molecules to the target, via which an electric charge can be delivered into the target to cause damaging effects. These can be used to destroy anything that conducts electricity better than the air or ground surrounding it. This works as follows:

A high-intensity train of picosecond laser pulses is used to create a powerful electromagnetic pulse around itself that strips electrons from air molecules, thereby creating a plasma channel through the air. Since the air is composed of neutral particles that act as insulators, LIPC is relatively a good conductor. A high-voltage current discharge is sent down this conducting filament to the target rather than arcing unpredictably through the air—a phenomenon similar to lightning that finds its way from clouds to ground via the path of least resistance.

Of these four categories, high-energy laser weapons have the greatest potential in the near term to become worthy of a potent weapon system.

High-power microwave technology, too has similar potential, but has not been funded as generously as high-energy laser weapon development programmes. LIPC has



Fig. 3: Concept of space based PBW in science fiction



Fig. 4: Another concept of space based PBW

target designation for munitions guidance and more will continue to improve as newer technologies evolve and develop, it is the use of lasers as weapon that is going to rewrite the military balance in the next 15 to 20 years.

Introduction of laser based DEWs is set to dramatically alter the war-fighting capabilities of nations by making possible execution of missions that would be extremely complex, if not impossible, to realise with conventional KEWs. These include ground based laser systems for disabling low Earth orbit satellites and destroying missiles,